

$$f(X) = \{x, y\} \notin \mathcal{V}_K$$

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$$f(\{c\}) = \{y\}$$

$$f(\emptyset) = \emptyset$$

Since $X \in \mathcal{T}_K$ but $f(X) \notin \mathcal{V}_K$

f is not a closed function.

(iii) $\{y\} \in \mathcal{V}$. But $f^{-1}(\{y\}) = \{b, c, d\} \notin \mathcal{T}$.

$\therefore f$ is discontinuous.

Definition: Two topological spaces $(X, \mathcal{T})$ & $(Y, \mathcal{V})$ are called

homeomorphic or topologically equivalent

if there exists a bijection $f: X \rightarrow Y$

s.t. both f & f^{-1} are continuous. Such

a function f is called a homeomorphism or a topological equivalence.

If $(X, \mathcal{T})$ & $(Y, \mathcal{V})$ are homeomorphic

then we write $X \cong Y$.

Proposition: The relation in any collection of topological spaces defined by " X is homeomorphic to Y " is an equivalence relation.

By the fundamental theorem on equivalence relations, any collection of topological spaces can be partitioned into classes of topologically equivalent spaces.

Definition: A property P of sets is called a topological property or a topological invariant if whenever a topological space has the property P then any topological space homeomorphic to that topological space has also the property P .

(Considering the fact that formally topology is the study of topological invariants this is an important definition!)

Convergent sequences

Definition: Let X be a nonempty set.

A function $x: \mathbb{Z}^+ \rightarrow X$ is called a sequence in X . Usually we denote this sequence by (x_n) or $\{x_n\}$.

Definition: Let (X, τ) be a topological space and (x_n) be a sequence of points in (X, τ) . We say that (x_n) converges to a point $x_0 \in X$ iff for every $U \in \tau$ with $x_0 \in U$ there exists an $n_0 \in \mathbb{Z}^+$ s.t.

$n \geq n_0 \Rightarrow x_n \in U$, i.e. if U contains almost all, that is all except

finitely many terms of the sequence then we say (x_n) converges to x_0 .

We write $\lim_{n \rightarrow \infty} x_n = x_0$ or $\lim x_n = x_0$ or

$x_n \rightarrow x_0$ as $n \rightarrow \infty$.

Example: Let $X \neq \emptyset$ and
 $\mathcal{C} = \{\emptyset\} \cup \{A \mid A \subseteq X, X \setminus A \text{ is countable}\}$.

Then $\mathcal{C}$ is a topology on X . It is called the cocountable topology or countable complement topology on X .

A proper subset K of X is closed $\Leftrightarrow$

$X \setminus K \in \mathcal{C} \Leftrightarrow K = (X \setminus (X \setminus K))$ is countable.

In other words a subset K of X is closed iff K is countable or $K = \emptyset$ or $K = X$.

Example: Let X be an uncountable set, $\mathcal{C}$ & $\mathcal{D}$ be the cocountable topology & the discrete topology on X . Recall that $\mathcal{D}$ is the finest topology on X . Hence $\mathcal{C} \subseteq \mathcal{D}$. But $\mathcal{D}$ is not coarser than $\mathcal{C}$. Hence if X is uncountable then the discrete topology & cocountable topology on X are different. For example take $X = \mathbb{R}$.

Let $A = \mathbb{R} \setminus \mathbb{Q}$. Because both $A = \mathbb{R} \setminus \mathbb{Q}$ and $\mathbb{R} \setminus A = \mathbb{Q}$ are subsets of $\mathbb{R}$, A is both $\mathcal{D}$ -open & $\mathcal{D}$ -closed.

Now because $\mathbb{R} \setminus A = \mathbb{Q}$ is countable $A \in \mathcal{T}$. But because $A = \mathbb{R} \setminus \mathbb{Q}$ is uncountable A is not $\mathcal{T}$ -closed.

Examples:

1. Let $(X, \mathcal{T})$ be the indiscrete topological space & (x_n) be a sequence of points in X . Show that (x_n) converges to every point x in X .

Solution: Let $x \in X$. Since $\mathcal{T}$ is the indiscrete or the trivial topology on X there is only one $\mathcal{T}$ -open set containing x . It is the set X itself. Because for any $n \geq 1$ $x_n \in X$.

$$\lim_{n \rightarrow \infty} x_n = x.$$

For example if $X = \mathbb{Z}^+$ & $\mathcal{T}$ is the trivial topology on X and (x_n) is any sequence of points in $\mathbb{Z}^+$ then $x_n \rightarrow 1, x_n \rightarrow 2, x_n \rightarrow 3, \dots$ as $n \rightarrow \infty$.

2. Let (x_n) be a sequence of points in a discrete topological space $(X, \mathcal{D})$.

Show that $\lim_{n \rightarrow \infty} x_n = x_0 \iff$ there exists a

positive integer n_0 s.t. $n \geq n_0 \implies x_n = x_0$,

i.e. (x_n) is of the form

$$(x_n) = (x_1, x_2, \dots, x_{n_0-1}, x_0, x_0, x_0, \dots)$$

Solution;

$\Rightarrow$ Assume that $\lim_{n \rightarrow \infty} x_n = x_0$. Since

$\{x_0\} \in \mathcal{D}$ with $x_0 \in \{x_0\}$ there exists

a positive integer n_0 s.t.

$n \geq n_0 \implies x_n \in \{x_0\}$, i.e.

$n \geq n_0 \implies x_n = x_0$.

Hence (x_n) is of the form

$$(x_n) = (x_1, x_2, \dots, x_{n_0-1}, x_0, x_0, x_0, \dots)$$

$\Leftarrow$ Let $U \in \mathcal{D}$ with $x_0 \in U$. Since

$x_n = x_0$ for all $n \geq n_0$, whenever

$n \geq n_0$ $x_n \in U$.

Thus $\lim_{n \rightarrow \infty} x_n = x_0$.

3. Let $(X, \mathcal{C})$ be a cocountable topological space & (x_n) be a sequence of points in X . Show that

$\lim_{n \rightarrow \infty} x_n = x_0 \iff$ there exists a positive

integer n_0 s.t. $n \geq n_0 \Rightarrow x_n = x_0$.

Solution:

$\Rightarrow$ Let $\lim_{n \rightarrow \infty} x_n = x_0$. Suppose, to the

contrary, that such a positive integer does not exist. Then the set

$$K = \{n \in \mathbb{Z}^+ \mid x_n \neq x_0\}$$

is a countably infinite. Now the set

$$F = \{x_n \mid n \in K\} = \{x_n \mid x_n \neq x_0\}$$

is countable (F may or may not be finite). Hence $U = X \setminus F \in \mathcal{T}$ and

$x_0 \in U$. Since $\lim_{n \rightarrow \infty} x_n = x_0$ there

exists a positive integer n_0 s.t.

$$n \geq n_0 \Rightarrow x_n \in U = X \setminus F. \quad (*)$$

Since K is an infinite set there exists a positive integer $n_1 \in K$ s.t.

$n_1 \geq n_0$. It follows from (*) that

$$n_1 \geq n_0 \Rightarrow x_{n_1} \in U = X \setminus F.$$

On the other hand because $n_1 \in K$ $x_{n_1} \in F$. This is a contradiction.

$\therefore (x_n)$ is of the form

$$(x_n) = (x_1, x_2, \dots, x_{n_0-1}, x_0, x_0, x_0, \dots).$$

$\Leftarrow$ Assume that (x_n) is of the

form $(x_n) = (x_1, \dots, x_{n_0-1}, x_0, x_0, \dots, x_0)$.

Let $U \in \mathcal{T}$ with $x_0 \in U$. Since $x_n = x_0$